

Price Accuracy and Resolution Capture in Prediction Markets

Anirudh Venkatesan

University of Virginia

June 29, 2026

Abstract

Prediction markets are praised as accurate and criticized as easy to manipulate. I argue these claims are not in conflict, because a prediction market has two separate parts. A trading layer sets the price, and a resolution layer decides the outcome after the event. The price corrects itself, because a wrong price is a chance to profit and other traders trade against it, but the resolution layer does not, because on platforms like Polymarket it is settled by a token-weighted vote, and a voter can vote for the outcome that pays their own position. Using 2,100 of the highest-volume resolved Polymarket markets, I find that the trading layer is well calibrated, with a regression of outcomes on prices that cannot be distinguished from perfect calibration. The documented capture cases instead cluster in markets with vague resolution rules, with one clear-rule case that points to a second channel through voting power. Because these prices are now used by news outlets and automated systems, a captured resolution is not just money moving between traders but a cost imposed on everyone who relies on the price.

Keywords: prediction markets, market design, information aggregation, oracles, blockchain, market manipulation.

JEL classification: D47, D82, G14, D72.

1 Introduction

A central question about prediction markets is whether their prices can be trusted as estimates of how likely future events are. Prediction markets allow participants to trade contracts that pay one dollar if an event occurs and nothing otherwise, so the price of a contract is commonly read as the market's probability that the event will happen. A large body of research argues that these prices combine scattered information well, partly because traders have to put money behind their views (Wolfers & Zitzewitz, 2004; Manski, 2006). Prediction-market prices have outperformed polls in several settings, and they are now cited by news organizations, integrated into financial data terminals, and used by automated systems as real-time probability estimates.

At the same time, decentralized prediction markets are frequently accused of manipulation. Between 2024 and 2026, as platforms such as Polymarket grew rapidly, several high-volume markets

settled in ways that appeared to contradict the underlying facts. The usual response treats accuracy and manipulation as competing answers to a single question about whether prediction markets work. In this paper, I argue that the question is better split into two.

A prediction market has a trading layer, in which participants buy and sell contracts until a price forms, and a resolution layer, in which a settlement process determines the outcome after the event so that winning contracts can be redeemed. The self-correcting feature usually credited to prediction markets is the idea that an informed trader can profit by pushing a wrong price back to its correct value. This is a feature of the trading layer. The resolution layer is not a trading process. On Polymarket, contested outcomes are settled by a token-weighted vote through the UMA optimistic oracle. No argument about arbitrage applies to a vote. The word “market” makes people assume the accuracy of the price also applies to the resolution stage, even though that stage works in a completely different way. I call the line between these two parts the resolution boundary. The main claim of this paper is that the feature making the trading layer reliable does not hold once you cross that line.

This point is not simply that prices self-correct, which is already well known. It rests on three claims. First, the self-correction result has a boundary, because resolution is decided by voting rather than trading. Second, the return to causing an error differs in sign across the two layers. In the trading layer, causing a price distortion is costly, because a trader who moves the price away from the truth loses at resolution. In the resolution layer, causing a wrong outcome can be profitable, because a voter can vote for the outcome that pays their own position, and no one can profit by trading against a vote. Third, the harm changes once the price is used outside the market. A captured resolution that used to just move money between traders now imposes a cost on everyone outside the market who uses the price.

I study these claims using 2,100 of the highest-volume resolved Polymarket markets. I find that the trading layer behaves as the optimistic literature predicts. Using prices recorded roughly one day before each market closed, contracts priced above 0.8 were realized 83 percent of the time and contracts priced below 0.2 were realized 7 percent of the time, and a regression of outcomes on prices cannot reject perfect calibration. I then construct an index of how subjective each market’s resolution rule is, and I show that the documented capture cases concentrate in the ambiguous region of this index. The one exception, a market on a Ukraine mineral agreement, has a clear resolution rule and was captured regardless, which I interpret as evidence of a second channel that operates directly through voting power.

2 Background

Polymarket is a decentralized prediction market deployed on the Polygon blockchain. Participants trade binary outcome tokens priced between zero and one dollar using a hybrid system with an off-chain order book and on-chain settlement. Resolution is handled separately from trading. An outcome is first proposed, a challenge window opens, and if the proposal is disputed the question is escalated to the UMA optimistic oracle, where holders of the UMA token vote on the correct

outcome with voting power proportional to the tokens they stake (UMA, 2024). Voters in the majority are rewarded and voters in the minority can be penalized, a design meant to make honest voting the natural choice, in the sense of Schelling (1960).

Two features of this design are central to the analysis. First, resolution is decided by a vote, not by trading. Second, nothing in the mechanism prevents a voter from holding a position in the market being resolved. Investigative reporting indicates that in most disputed markets the majority of votes came from a small number of large wallets, that a majority of active voters were linked to live trading positions, and that a meaningful share of disputes included a voter with a position in the disputed market (WSJ, 2026). As prediction-market prices have been adopted by mainstream media and automated systems, the consequences of an incorrect resolution have extended beyond the participants in a given market.

The idea that prediction-market prices can be read as probabilities is developed by Wolfers & Zitzewitz (2004) and Manski (2006). It builds on Hayek (1945), who described prices as a way to combine scattered information, and on Grossman & Stiglitz (1980), who showed that gathering information is costly and must be rewarded. The literature on manipulation finds that moving prices is difficult to do profitably. Hanson & Oprea (2009) show that a manipulator can in fact improve accuracy by subsidizing informed traders, and experimental and historical studies generally find that attempts to move prices fail. The main documented anomaly in the price itself is the favorite-longshot bias (Thaler & Ziemba, 1988; Snowberg & Wolfers, 2010). There is one important exception. Kumar & Seppi (1992) show that distorting a price becomes profitable when it also pays off somewhere outside the market, such as a separate cash-settlement position. The resolution vote functions as exactly such an outside payoff. Despite the size of the price-efficiency literature, the settlement layer of decentralized prediction markets has received little attention from economists. The closest study is Abid (2026), which documents that the average price is corrected by counterparty incentives while the resolution layer is not robust to concentration of capital. I build on that work by stating the difference between the two layers as two clear propositions, and by testing the role of vague rules using a measure I apply to a large sample of markets.

3 A Two-Layer Framework

3.1 Setup

Consider a binary contract that pays one dollar if event E occurs and nothing otherwise. Two mechanisms act in sequence. The trading layer produces a price $p \in [0, 1]$ through continuous trading until the market closes. The resolution layer applies a settlement process R that determines an outcome $\hat{y} \in \{0, 1\}$ and pays contracts accordingly. The value of a contract is p before close and $\hat{y}$ after. The price tells us something about the event only if R is honest, meaning $\hat{y}$ equals the true outcome y .

3.2 The trading layer self-corrects

Suppose a trader moves p away from the probability that other informed traders assign to E . Any trader who thinks the price is wrong expects to profit by trading against it, so informed money enters until the price returns to its earlier level. The expected return to causing a price distortion Δ is therefore

$$\Pi^{\text{trade}}(\Delta) \approx -c(\Delta) - \mathbb{E}[\text{loss at honest resolution}] < 0,$$

where $c(\Delta) \geq 0$ is the cost of moving the price. The return is negative, so price distortions are self-limiting. This is consistent with Hanson & Oprea (2009) and with the broader evidence that price manipulation is difficult. It also fits the most discussed example of apparent manipulation, the large trader who moved the 2024 presidential price. On closer inspection, that trader held an informed and contrarian view and profited from a correct forecast, not from manipulation.

3.3 The resolution layer does not

Resolution is decided by a token-weighted vote, and three features prevent the self-correction argument from applying.

First, a vote is not a trade. There is no counterparty who can profit by buying an incorrect vote and selling it back, so the arbitrage force that disciplines the price has no analogue at the resolution stage.

Second, the focal point fails when rules are ambiguous. The optimistic oracle is a Schelling mechanism in which voters are expected to converge on the truth as a unique focal point (Schelling, 1960). This works for clean and verifiable facts. When a resolution rule is ambiguous, for example when it turns on whether a public figure wore a suit, on what counts as credible reporting, or on whether a sale occurred in a particular month, there are several defensible focal points, and the largest concentration of voting power selects among them.

Third, voters may hold positions in the market they adjudicate. The privately optimal vote is then toward the voter's own position rather than toward the truth, and token-weighting gives the most exposed participants the most influence. The expected return to causing a misresolution is

$$\Pi^{\text{resolve}}(\Delta) \approx +V_{\text{position}} - b,$$

where V_{position} is the value of the position protected by a favorable vote and b is the bond or penalty at risk. Whenever $V_{\text{position}} > b$, the return is positive.

3.4 Two propositions

The argument can be stated compactly.

Proposition 1 (Sign reversal at the resolution boundary). *The expected return to causing a distortion is negative in the trading layer, $\Pi^{\text{trade}}(\Delta) < 0$, and positive in the resolution layer*

whenever the protected position exceeds the bond at risk, $\Pi^{\text{resolve}}(\Delta) > 0 \iff V_{\text{position}} > b$. Distortions are therefore self-limiting in price and self-reinforcing in resolution.

Proposition 2 (Ambiguity lowers the cost of capture). *Let a resolution rule admit $k \geq 1$ defensible focal points. Under a Schelling vote, the weighted stake required to move the settled outcome to a preferred focal point is decreasing in k . When $k = 1$, capture must overcome the honest majority. When $k > 1$, a plurality of weighted stake is sufficient. The incidence of capture is therefore increasing in the ambiguity of the rule and in the concentration of voting power.*

Proposition 1 states the asymmetry. Proposition 2 generates the empirical prediction tested below, namely that capture should concentrate in markets with ambiguous resolution rules, with concentration of voting power providing a second channel when ambiguity is absent.

3.5 Why the stakes have grown

When a prediction-market price gets little attention and only matters inside the market, a captured resolution simply moves money between traders. Once the price is used outside the market, as a measure of sentiment, as an input to news reporting, or as a value retrieved by automated systems, a captured resolution records an incorrect outcome in a figure that others rely on. This is an information externality, because people who use the price cannot see whether a given resolution was disputed or how it was decided. It also fits the general pattern that a measure becomes less reliable as people start to rely on it, and here the point of failure is the resolution boundary.

4 Data

I use the 2,100 highest-volume resolved markets retrieved from Polymarket’s public Gamma application programming interface, which does not require authentication. The sample spans November 2020 through January 2027. Total traded volume is approximately 37.2 billion dollars, the median market volume is 7.6 million dollars, and the largest market, the 2024 United States presidential election, traded 1.53 billion dollars. For each market I observe the question, the full resolution rule, the settled outcome, volume, and timing. The realized base rate of “Yes” outcomes is 26.6 percent, which reflects the prevalence of contracts on whether a specific event will occur. Resolution rules average 737 characters, and 49.5 percent contain at least one subjective resolution term. Table 1 summarizes the sample.

For the calibration test I additionally retrieve, from Polymarket’s public order-book price history, each binary market’s implied probability roughly 24 hours before it closed. Usable pre-close prices are available for 201 markets. The order-book history does not extend to markets that resolved before 2022, which are therefore excluded from the calibration sample.

Data limitations. I test two predictions of the framework using public data, namely trading-layer calibration and the ambiguity channel. The vote-level records needed to measure which wallets voted and what positions they held, which would provide a direct test of the conflict channel, require

Table 1: Dataset descriptives.

Markets	2,100
Date range (resolution)	Nov 2020 to Jan 2027
Total traded volume	\$37.2 billion
Median market volume	\$7.6 million
Largest market	\$1.53 billion
Realized “Yes” base rate	26.6%
Mean resolution-rule length	737 characters
Markets using subjective resolution language	49.5%
Calibration subsample (retrievable pre-close price)	201

data access and on-chain position tables that were not freely available for this draft. I therefore treat the conflict channel through documented cases and prior reporting rather than measuring it directly. I state this boundary rather than work around it.

5 Results I: Calibration of the Trading Layer

I first verify the premise that the trading layer produces meaningful probabilities. For each market with retrievable order-book history I take the implied probability roughly 24 hours before close and compare it to the realized outcome. If prices are honest probabilities, contracts priced at q should occur a fraction q of the time.

Figure 1 reports the result. The binned calibration points lie close to the 45-degree line. Contracts priced near 0.02 occurred 5 percent of the time, those near 0.15 occurred 16 percent of the time, and those near 0.98 occurred 94 percent of the time. The Brier score of the market price is 0.145, compared with 0.236 for a forecast that always predicts the sample base rate, an improvement of 39 percent. Contracts priced above 0.8 were realized 83 percent of the time, and contracts priced below 0.2 were realized 7 percent of the time. This establishes that the price layer is healthy, so that the resolution failures documented in the next section cannot be attributed to a generally inaccurate market.

I next test the favorite-longshot bias, in which longshots are overpriced and favorites underpriced. I regress the realized outcome on the pre-close price, $y = a + bp + \varepsilon$, where perfect calibration implies $a = 0$ and $b = 1$. The estimates are $a = 0.05$ with a standard error of 0.04 and $b = 0.94$ with a standard error of 0.08, and the t -statistic for the hypothesis $b = 1$ is -0.78 . The point estimates lean in the direction of the classic bias, since a slope below one and a positive intercept imply mild overpricing of longshots, but the deviation is not statistically distinguishable from perfect calibration at this sample size (Figure 2). I read this as evidence that any favorite-longshot bias on Polymarket is at most mild and that the trading layer is close to efficient.

6 Results II: The Resolution Layer and the Ambiguity Channel

Proposition 2 predicts that, holding the trading layer fixed, resolution failures should concentrate in markets whose rules admit several defensible focal points. To test this, I construct an index of

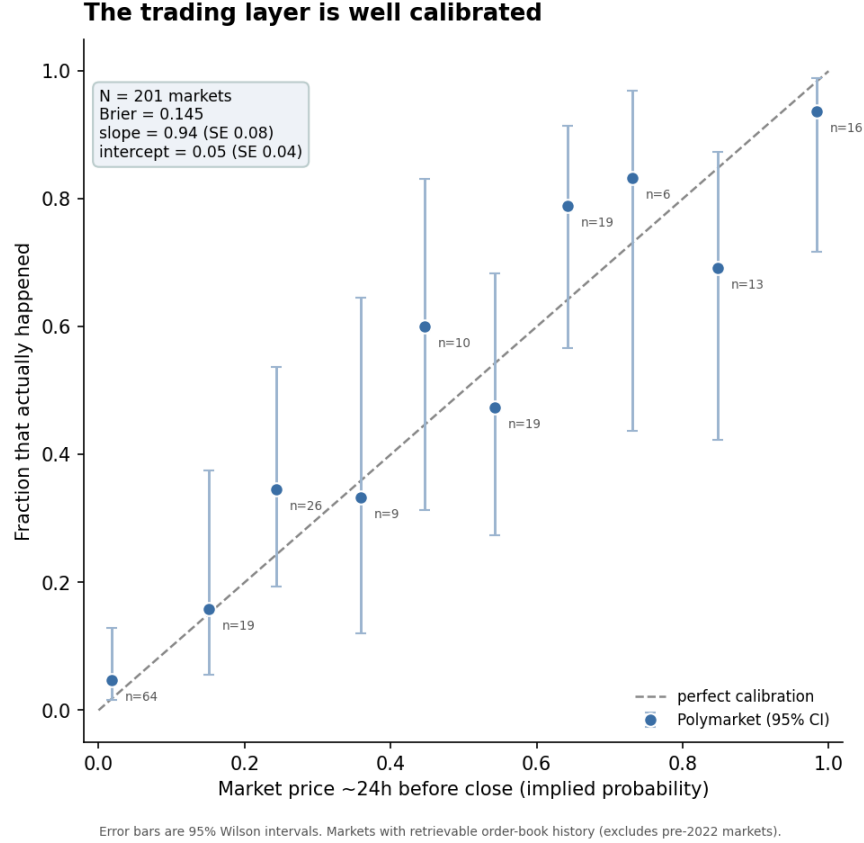


Figure 1: Calibration of pre-close prices against realized outcomes for the 201 markets with retrievable order-book history. Point size is proportional to the number of markets in each bin. Proximity to the dashed line indicates accurate prices.

resolution-rule ambiguity from the text of each rule:

$$\text{Ambiguity} = (\text{subjective terms}) - 0.5 (\text{objective anchors}) + 0.25 (\text{edge-case clauses}),$$

where subjective terms include language such as credible, consensus, widely reported, discretion, and judgment; objective anchors include explicit sources, web addresses, official releases, and numeric thresholds; and edge-case clauses count conditional constructions. The index is intentionally simple and reproducible rather than a black box.

I then locate the four documented governance-attack markets that are present in the sample, namely the Ukraine mineral-deal market, the Zelensky-suit market, the MicroStrategy-Bitcoin market, and the UFO-declassification market, and I compare their ambiguity to the full distribution (Figure 3). Three of the four lie in the ambiguous region. The Zelensky-suit market is at the 85th percentile, the MicroStrategy market at the 82nd, and the UFO market at the 73rd. The median disputed case is at the 77.5th percentile, against 50 for a market drawn at random. To assess whether this clustering could arise by chance, I draw four markets at random 200,000 times and compute the median ambiguity percentile of each draw. The probability that a random set of four

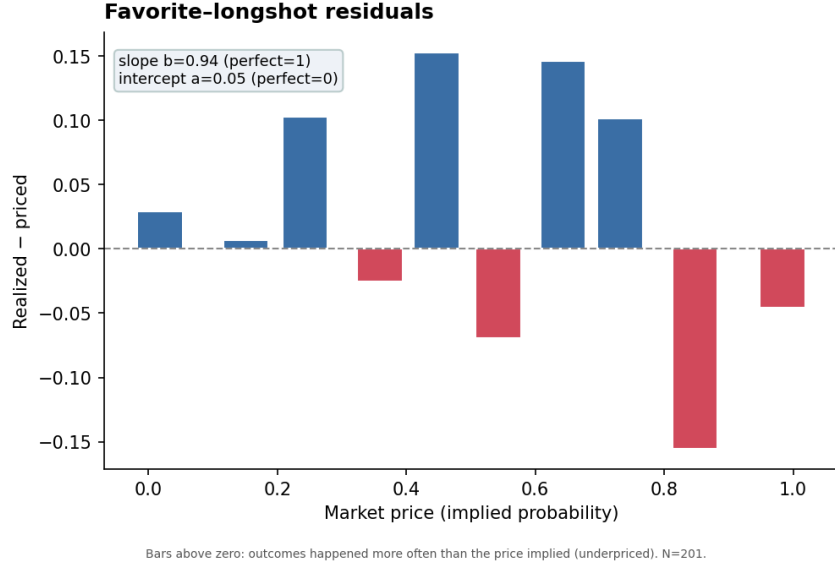


Figure 2: Residuals, defined as realized minus priced, by price bin. Bars above zero indicate outcomes that occurred more often than the price implied. The pattern is weak and not statistically significant ($N = 201$).

markets has a median ambiguity percentile of at least 77.5 is 0.068, and the probability that at least three of four random markets fall above the 73rd percentile is 0.077. The clustering is therefore marginally significant rather than merely suggestive, though the small number of documented cases means this evidence should be read as indicative. The fourth case, the Ukraine mineral-deal market, is at the 13th percentile. It has a clearly phrased rule and was captured regardless. I do not read this as evidence against the framework but as direct evidence of the second channel in Proposition 2, in which capture operates through voting power rather than through ambiguity. Table 2 summarizes the four cases.

The ambiguity of resolution rules also varies by topic (Figure 4). Markets on economic indicators, which typically resolve against numeric releases such as inflation prints and interest-rate decisions, are the most objective. Markets on cryptocurrency, which often resolve against exchange or on-chain data, are next. Markets on politics and geopolitics, which is the category that contains all four documented capture cases, are considerably more subjective, with 68 percent using discretionary resolution language. This is consistent with the framework, since the markets most exposed to capture are those whose questions resist a single verifiable answer.

Table 2: The four documented resolution-capture cases.

Market	Stakes	Ambiguity pct.	Primary channel
MicroStrategy Bitcoin (May)	\$376M	82nd	Rule ambiguity
Zelensky suit	\$242M	85th	Rule ambiguity
UFO declassification	\$16.7M	73rd	Rule ambiguity
Ukraine mineral deal	\$7.1M	13th	Voting-power override

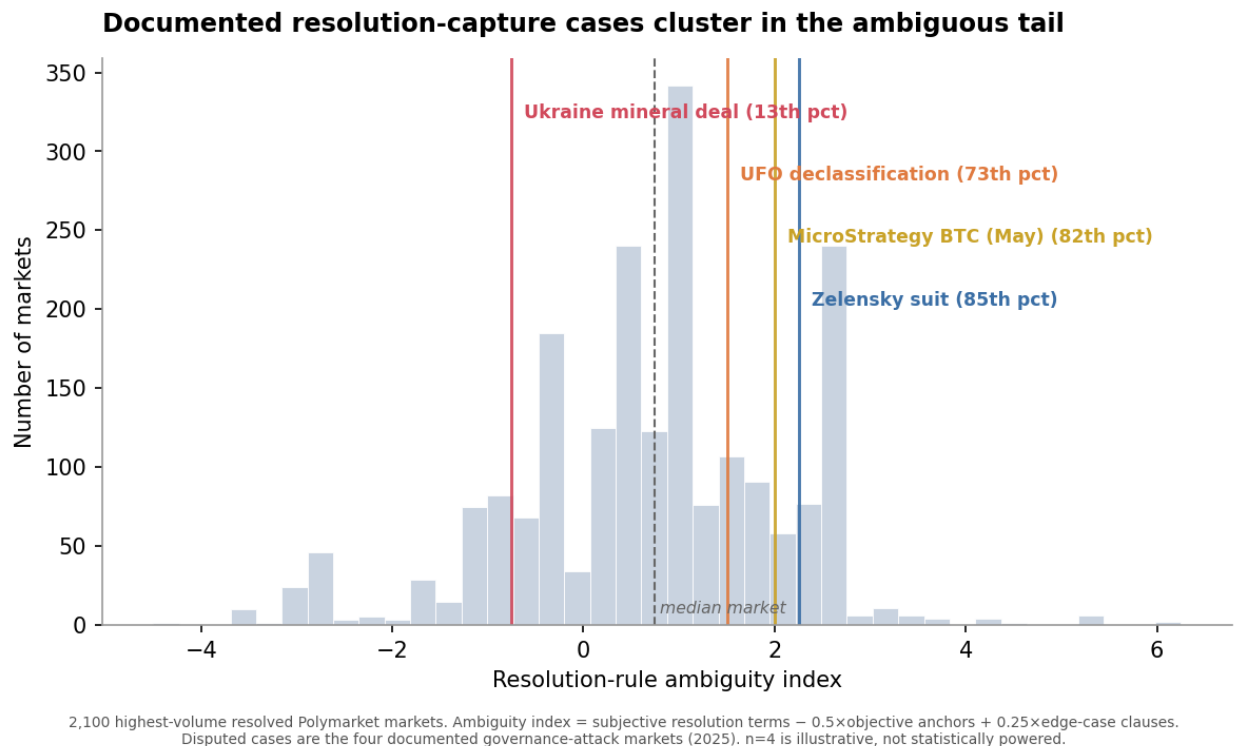


Figure 3: Distribution of the resolution-rule ambiguity index across 2,100 markets, with the four documented governance-attack markets marked. Three of the four lie in the ambiguous region, and the Ukraine mineral-deal market is a low-ambiguity outlier. The four documented cases are illustrative rather than a statistically powered sample.

I note the limitations of this exercise directly. There are only four documented capture cases, so the comparison is illustrative rather than a powered statistical test. The ambiguity index is a heuristic. Markets that were disputed but not captured are not yet labeled. What the exercise establishes is that the mechanism the framework identifies, namely ambiguity as the channel through which concentrated capital selects the outcome, is present in the cases available, and that the single exception is itself informative about the second channel.

7 Case Studies

The following episodes, drawn from public reporting, illustrate the resolution layer failing in ways the trading layer does not.

In the Ukraine mineral-deal market in March 2025, a market on whether Ukraine would agree to a United States mineral deal moved sharply and resolved “Yes” despite the absence of an agreement. Reporting attributes the outcome to a single large holder who cast roughly a quarter of the votes in the dispute round, and the platform acknowledged that the result was incorrect (The Block, 2025). The rule was not ambiguous, which makes this a clear case of capture through voting power.

In the Zelensky-suit market in mid-2025, a market exceeding 200 million dollars on whether

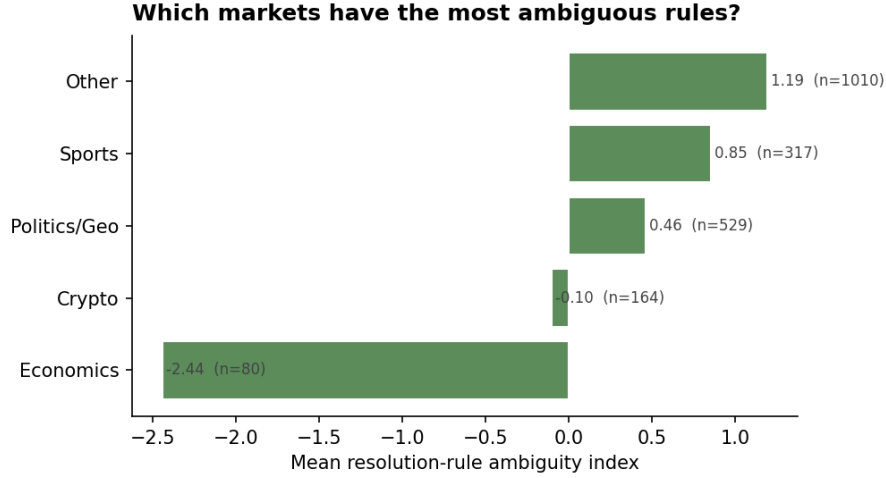


Figure 4: Mean resolution-rule ambiguity by market category. Politics and geopolitics markets, where the documented capture cases occur, rely heavily on subjective resolution language.

the Ukrainian president had worn a suit resolved against the apparent evidence. The ambiguity of what counts as a suit created the multiple focal points described in the framework (Cryptopolitan, 2025).

In the MicroStrategy market in 2026, a market of tens of millions of dollars turned on whether a disclosed sale in May required public disclosure within the month or only execution on-chain, an ambiguity in the rule itself (The Defiant, 2026).

In the UFO-declassification market in 2025, a market resolved “Yes” on whether files had been declassified despite no documents being released, a result that turned on what counts as credible reporting (CryptoSlate, 2025).

In each case the trading price had been informative until the resolution stage. The failure is located at the resolution boundary.

8 Robustness

I check that the two main results do not depend on specific modeling choices.

Calibration is robust to the timing of the price. The calibration result uses the price recorded roughly 24 hours before a market closed. Repeating the regression with the price recorded roughly seven days before close gives an almost identical picture: an intercept of 0.048 with a standard error of 0.047 and a slope of 0.954 with a standard error of 0.094, against 0.049 and 0.936 for the 24-hour price. The Brier score changes only slightly, from 0.145 to 0.152. The conclusion that the trading layer is well calibrated does not depend on the exact moment at which the price is sampled.

The ambiguity result is robust to the construction of the index. The ambiguity index combines subjective terms, objective anchors, and edge-case clauses. To check that the result is not an artifact of these weights, I recompute each market’s ambiguity percentile under two simpler definitions: the count of subjective terms alone, and subjective terms minus objective anchors. Table 3 reports the percentiles of the four documented cases under each definition. The three ambiguous cases remain in the upper part of the distribution under every definition, and the Ukraine mineral-deal market remains a low-ambiguity outlier under every definition. The pattern does not depend on the particular form of the index.

Table 3: Ambiguity percentile of the four documented cases under three index definitions.

Market	Main index	Subjective only	Subjective minus objective
Zelensky suit	85	75	97
MicroStrategy Bitcoin (May)	82	75	88
UFO declassification	73	75	63
Ukraine mineral deal	13	25	25

Ruling out a thin-market explanation. A natural alternative is that the captured markets were simply thin markets dominated by a few traders, so that the same concentration drove both trading and voting. Using holder data from Polymarket’s public Data API, I find that the four captured markets were among the most widely held in the sample, each with holdings spread across the maximum number of holders the query returned. This does not measure who voted in the disputes, which would require vote-level records on a separate chain, but it rules out the explanation that capture reflected thin or inactive markets. The trading side of these markets was broad, which localizes the failure to the resolution layer.

9 The Cost of a Captured Resolution

The framework so far explains why the resolution layer can be captured. This section asks what that capture costs once the price is used outside the market. I present a simple model to make the externality argument precise rather than verbal.

Consider a single binary question with true outcome $y \in \{0, 1\}$. A manipulator holds a position of size V in the market and can force the resolution to a preferred outcome by paying a capture cost κ , where κ depends on the ambiguity of the rule and the concentration of voting power as described in Proposition 2. The manipulator captures the resolution whenever

$$V > \kappa.$$

Inside the market alone, capture is a transfer. The manipulator gains V and the counterparties lose V , so the net effect on traders sums to zero apart from the deadweight cost κ .

Now suppose the settled outcome is also consumed outside the market. Let there be n outside

agents who cannot observe the dispute and who treat the settled value $\hat{y}$ as the truth. Each agent j takes an action and suffers a loss ℓ_j when the recorded outcome is wrong, so the external damage from a captured resolution is

$$D = \sum_{j=1}^n \ell_j.$$

The manipulator does not pay D . The private condition for capture remains $V > \kappa$, but the socially efficient condition is $V > \kappa + D$, because society bears the external damage as well. Capture is therefore privately profitable yet socially harmful for the entire range

$$\kappa < V < \kappa + D.$$

The size of this region grows with n , the number of outside users of the price. This is the formal content of the claim that the stakes rise as adoption rises. When the price was a low-attention figure, n was small, D was negligible, and private and social incentives nearly coincided. As news organizations and automated systems began to treat the price as a fact, n and therefore D grew, and a widening set of captures became privately profitable while socially destructive.

Two implications follow. First, the harm is an externality in the standard sense: the manipulator’s decision ignores D , so the market produces too many captured resolutions relative to the efficient level. Second, the externality is increasing in the very feature that makes prediction markets valuable, namely their adoption as public information. A market that no one relies on cannot impose this cost, and a market that everyone relies on imposes it most. The policy tools discussed in the next section can be read as attempts to raise κ (by making capture harder) or to reduce D (by signaling to outside users when a resolution was contested).

10 Discussion and Policy

If the problem is located in the resolution layer, the available remedies are matters of mechanism design, and each involves a trade-off. Deterministic settlement, in which an outcome is resolved from verifiable data with no vote, removes discretion but applies only to facts that can be verified by machine, and so it defines the set of questions that can be listed responsibly. Recusal rules that bar voting by holders of a position address the conflict channel directly but are difficult to enforce under pseudonymity. Restricting voting to a whitelist of approved voters reduces noise but re-centralizes the process and does not help if the same large holders are approved. Replacing the external oracle with a token internal to the platform relocates the concentration problem rather than removing it. In the language of the previous section, deterministic settlement and recusal rules raise the capture cost κ , while clearer disclosure of contested resolutions lowers the external damage D by warning outside users not to treat a disputed outcome as settled fact. The general point is that for ambiguous questions one cannot have both full decentralization and reliable settlement, because discretion must reside somewhere, and wherever it resides it can be captured. The implication is that platforms should match the questions they list to the settlement technology available, listing

only questions that can be settled reliably. Because these prices now function as a public input, the case for treating settlement integrity as a regulated function is also stronger than the price-efficiency literature alone would suggest.

11 Conclusion

The apparent conflict between the claim that prediction markets are accurate and the claim that they are manipulated is resolved once the platform is separated into two components. The trading layer self-corrects because a mispriced contract is a profit opportunity. The resolution layer does not self-correct because a misresolution is a payout to the voter who causes it. Evidence from 2,100 markets is consistent with this account. The trading layer is well calibrated, with a regression of outcomes on prices that cannot reject perfect calibration, and the documented capture cases concentrate in markets with ambiguous resolution rules, with a low-ambiguity outlier that points to a second channel operating through voting power.

This paper has several limitations. The calibration sample is restricted to the 201 markets with retrievable order-book history, which excludes markets that resolved before 2022. The ambiguity index is a simple heuristic measured over rule text. The conflict-of-interest channel is documented through reporting rather than measured, because the vote-level and position data needed to measure it were not freely accessible. There are only four documented capture cases, so the ambiguity comparison is illustrative.

These limitations also point to a clear path forward. Obtaining vote-level and position data would allow the conflict channel to be measured directly rather than inferred from reporting, and labeling markets that were disputed but not captured would turn the ambiguity comparison into a powered test. Beyond these extensions, the two-layer view suggests a broader agenda. As prediction-market prices are adopted more widely as public information, the value of getting settlement right grows, and the design of the resolution layer, rather than the accuracy of the price, becomes the central question for whether these markets can be trusted. The trading layer already works. The task ahead is to build a resolution layer that deserves the same confidence.

A Construction of the Ambiguity Index

The ambiguity index is computed from the natural-language resolution rule of each market as

$$\text{Ambiguity} = (\text{subjective terms}) - 0.5 (\text{objective anchors}) + 0.25 (\text{edge-case clauses}).$$

Subjective terms are counts of language that requires interpretation, including the stems and phrases: credible, consensus, widely reported or accepted, generally, reasonable, discretion, ambiguous, unclear, interpret, spirit of, judgment, subjective, deemed, good faith, resolver, and common sense. Objective anchors are counts of language that points to a verifiable source, including: according to, official, web addresses, named wire services, the Federal Reserve, announcements,

press releases, certified or final results, explicit numeric thresholds, closing prices, and on-chain or blockchain references. Edge-case clauses count conditional constructions using if and however. All counts are case-insensitive. The index is intentionally transparent so that it can be reproduced exactly from the resolution text, and Section 8 shows that the main result holds under simpler definitions that drop the weights.

B Data and Code Availability

All data were collected from public Polymarket endpoints that require no authentication: the Gamma API for market metadata and resolution rules, the CLOB price-history endpoint for pre-close prices, and the Data API for holders. The collection scripts and analysis code are released alongside this paper.

References

- Abid, M. N. (2026). The Polymarket Paradox: Manipulation, Whale Concentration, and Predictive Accuracy. Working paper, SSRN.
- Grossman, S. J., & Stiglitz, J. E. (1980). On the Impossibility of Informationally Efficient Markets. *American Economic Review*, 70(3).
- Hanson, R., & Oprea, R. (2009). A Manipulator Can Aid Prediction Market Accuracy. *Economica*, 76(302).
- Hayek, F. A. (1945). The Use of Knowledge in Society. *American Economic Review*, 35(4).
- Kumar, P., & Seppi, D. J. (1992). Futures Manipulation with Cash Settlement. *Journal of Finance*, 47(4).
- Manski, C. F. (2006). Interpreting the Predictions of Prediction Markets. *Economics Letters*, 91(3).
- Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.
- Snowberg, E., & Wolfers, J. (2010). Explaining the Favorite-Longshot Bias: Is It Risk-Love or Misperceptions? *Journal of Political Economy*, 118(4).
- Thaler, R. H., & Ziemba, W. T. (1988). Anomalies: Parimutuel Betting Markets. *Journal of Economic Perspectives*, 2(2).
- UMA Protocol (2024). How Does UMA’s Oracle Work? Documentation.
- Wolfers, J., & Zitzewitz, E. (2004). Prediction Markets. *Journal of Economic Perspectives*, 18(2).
- The Block (2025). Polymarket reports governance attack by UMA whale on bet resolution.
- Cryptopolitan (2025). Polymarket community protests oracle vote by UMA whales.

The Defiant (2026). Polymarket dispute over MicroStrategy’s May Bitcoin sale and UMA token-voting oracle.

CryptoSlate (2025). Polymarket credibility concerns after whales force a “Yes” UFO resolution without evidence.

Wall Street Journal (2026). Investigation of UMA dispute voting and Polymarket position overlap.